

Elicited Confidence Is Not a Safety Signal: Task-Dependent Reliability of Verbalized Uncertainty in Clinical Language Models

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Abstract—Confidence-gated deferral—routing low-confidence model outputs to a clinician—is the most widely proposed safety mechanism for deploying large language models (LLMs) in clinical decision support. Its validity rests on an untested premise: that elicited confidence discriminates correct from incorrect clinical outputs. We evaluate that premise across 25,485 model episodes spanning five LLMs from two vendors, six benchmarks, two answer formats, and both static and interactive consultation settings. Discrimination is strongly task-dependent. On standard clinical items, elicited confidence separates correct from incorrect answers (AUROC 0.67–0.70); at the capability frontier it collapses to chance (0.48–0.54), and this collapse is not an artifact of base rate—after resampling standard items to the frontier accuracy, AUROC is unchanged (0.698/0.703). Critically, on a probe of 99 items carrying can’t-miss diagnoses, discrimination degrades beyond what difficulty alone predicts (Δ AUROC -0.082 and -0.063 versus accuracy-matched controls), and confidence carries almost no information about whether the can’t-miss diagnosis was itself missed (AUROC 0.532/0.567); deferring the least-confident 22% of these cases reduces the missed-red-flag rate by only 3 percentage points. In contrast, when the model gathers its own evidence through multi-turn consultation, discrimination rises to 0.75/0.80 and deferring 18.5% of cases raises accuracy of the retained set by 8.3 points. Confidence elicited on static benchmarks therefore neither predicts nor bounds the reliability of confidence in interactive deployment, and confidence-gated deferral fails precisely on the cases it is intended to catch.

Index Terms—Clinical decision support, large language models, uncertainty quantification, selective prediction, patient safety, model calibration.

I. Introduction

LARGE language models are entering clinical decision support through triage assistants, differential-diagnosis aids, and documentation tools. Because such systems will be wrong some of the time, essentially every proposed deployment architecture includes an abstention pathway: the model reports a confidence, and outputs below a threshold are escalated to a clinician. This design is attractive because it requires no access to model internals—the confidence is simply elicited in natural language—and because it maps onto existing clinical triage workflows.

The design presumes that elicited confidence is informative about correctness. In the general machine-learning literature this presumption has been repeatedly questioned:

verbalized confidence is systematically overconfident [3], [2], its structure across models is dominated by a single shared item-difficulty factor rather than by any model-specific self-knowledge [14], and overconfidence has been traced to a failure to condition the stated confidence on the model’s own answer [6]. What that literature does not address, because it lacks clinical risk labels, is the question a clinical deployment actually turns on: does confidence remain informative on the cases where being wrong is most consequential?

A safety mechanism that works on average but fails on high-acuity presentations is worse than no mechanism, because it produces false assurance exactly where assurance is unwarranted. We therefore evaluate elicited confidence under three conditions that matter clinically and are absent from prior general-domain work: (i) at the model’s capability frontier, (ii) on cases carrying time-critical can’t-miss diagnoses, and (iii) in interactive consultation, where the model must decide for itself what history to take and what tests to order.

A. Contributions

- We show that the discriminative power of elicited confidence is task-dependent rather than a fixed model property, ranging from chance (0.48) to strong (0.80) within the same model, and we establish that the frontier collapse survives accuracy matching and is therefore not a base-rate artifact.
- We construct RedFlag-99, a probe of 99 clinical items carrying can’t-miss diagnoses across 44 conditions, built by a pre-registered sentinel-feature filter rather than by gold-label matching—which recovers only 41 items from 7,956—and show that confidence degrades on these items beyond what difficulty predicts.
- We show that confidence is near-uninformative about the clinically decisive endpoint—whether the can’t-miss diagnosis entered the differential at all (AUROC 0.532/0.567)—and quantify the resulting futility of confidence-gated deferral by decision-curve analysis.
- We show that the same models, on the same case material, yield substantially more informative confidence when they gather evidence interactively (AUROC 0.75/0.80), establishing that static benchmark estimates of confidence reliability do not transfer to interactive deployment in either direction.

II. Related Work

A. Eliciting and calibrating confidence in language models

Kadavath et al. [2] established that models can, under favorable conditions, self-evaluate proposed answers. Black-box elicitation was then systematized by Xiong et al. [3], who factored the problem into prompting strategy, sampling, and aggregation and benchmarked the combinations across five datasets; they report that every method examined degrades on tasks demanding professional knowledge—the regime this paper isolates. Subsequent work has largely pursued mitigation. Self-evaluation can be adapted for selective prediction [4]; calibration can be improved by self-generated distractors [7], by aligning verbalized with internal confidence [8], [10], by order-aware alignment [9], or by self-calibration at test time [5]. ADVICE [6] identifies answer-independence—the failure to condition the stated confidence on the model’s own answer—as a primary driver of overconfidence and corrects it by fine-tuning. Surveys and empirical audits confirm that overconfidence persists across current models [11], [12], [21].

B. Is elicited confidence metacognition?

A separate line asks what elicited confidence measures rather than how to fix it. MIRROR [13] benchmarks metacognitive calibration hierarchically; other work probes introspection directly [15] or attempts to steer it [16]. Most relevant to us, Moran et al. [14] decompose binary confidence judgements from 20 models across six benchmarks and find the cross-model confidence matrix approximately rank-one: models share an item-level difficulty axis and differ mainly in decision threshold, with the confidence–performance relationship vanishing once items all models agree on are removed. They report no evidence of individuated metacognition in any domain tested. A complementary result shows that confidences do not govern behavior: models act against their own stated confidences in agentic settings [17]. Our general-domain observations—strong cross-model agreement in confidence, near-zero within-model correlation with correctness—reproduce this picture, and we treat it as established background rather than as a contribution. What this line cannot address, lacking clinical risk labels, is whether confidence remains informative where error is most consequential.

C. Behavioral alternatives to verbalized confidence

Because stated confidence is unreliable, many systems substitute behavioral signals: answer consistency across samples [18], conformal procedures built on self-consistency sampling [19], or reasoning-conditioned estimates [20]. Selective prediction with such signals has been studied for code [22] and, in the medical setting, for vision-language triage [33]. We evaluate the two cheapest behavioral alternatives—sampling self-consistency and cross-model agreement—and find neither recovers discrimination at the capability frontier.

D. Test-time compute and its effect on confidence

Medical test-time scaling has been studied on the ascending half of the compute curve [25], [26]. Two results bear directly on our design. CDUR [20] reports that increasing chain-of-thought budget can induce overconfidence non-monotonically, and Hu et al. [27] show that enabling thinking shifts the composition of errors while aggregate accuracy barely moves. We therefore vary the effort parameter as a manipulation and verify that our conclusions hold at every operating point rather than at one.

E. Uncertainty and safety in clinical language models

Confidence in clinical LLMs has been examined by specialty [28] and by prompting strategy [29], generally reporting aggregate calibration on benchmark question sets. SaFE-Scale [30] shows that clinical safety and accuracy follow different scaling laws across deployment conditions including inference-time compute, and that consequential errors concentrate in a small subset of items—motivating our stratification by clinical risk rather than by average performance. Failure-mode analyses of clinical agents identify information-seeking failure, anchoring, and premature closure [31], [35], [34], and ResidencyRL [32] establishes missed-red-flag rate as a headline safety endpoint; we adopt that endpoint and ask whether elicited confidence predicts it. Multi-agent deliberation has been proposed to improve clinical reasoning [37], [36], and long-context clinical benchmarks expose anchoring to early interpretations [38]. Interactive clinical benchmarks [1] provide the consultation modality but have not been used to evaluate confidence, which is the gap our interactive experiments fill.

III. Materials and Methods

A. Models

Five models from two vendors spanning three capability tiers were evaluated: gpt-5.4, gpt-5.4-mini, gpt-5.4-nano, gpt-4o-mini (non-reasoning), and claude-sonnet-5. Including a non-reasoning model tests whether the phenomenon depends on explicit reasoning machinery. All calls were made through vendor APIs between 2026-08-28 and 2026-08-29; every response, token count, and latency was logged and cached.

B. Datasets

Clinical. MedAgentsBench [23] provides a hard subset (test_hard, 894 items over 10 sources) on which contemporary models sit near chance, and a standard-difficulty split from the same sources; using both isolates difficulty from corpus. AgentClinic [1] provides 214 OSCE vignettes with free-text gold diagnoses, used both as static vignettes and as interactive consultations. MedCalc-Bench [24] provides 1,100 procedural calculation items with an acceptance band and a ground-truth formula, permitting separation of execution slips from formula errors.

Non-clinical controls. MMLU-Pro, ARC-Challenge, and GSM8K were included as independent corpora spanning a wide accuracy range and two answer formats.

C. The RedFlag-99 probe

Filtering benchmark items by gold diagnosis recovers few high-acuity cases, because in exam-style items the gold answer is typically a management step: matching a can’t-miss diagnosis list against gold labels returned 41 items from 7,956 clinical items. We therefore filter on sentinel features in the stem. Forty-eight regular expressions encode presenting features that oblige exclusion of a time-critical diagnosis (thunderclap onset, pain out of proportion, saddle anesthesia, tearing chest pain with a pulse differential, and others), compiled from standard emergency-medicine can’t-miss lists. This is a mechanical criterion and requires no clinical judgement at filter time.

The filter returned 164 candidates. An automated annotator not among the evaluated models confirmed 105 as genuinely obliging exclusion of a can’t-miss diagnosis (yield 64%) and supplied an alias set for machine-checkable scoring; labels were then mapped onto a closed vocabulary (100% mapped, 46 conditions) and screened mechanically for fabricated sentinel quotes, alias/condition mismatch, and duplicates (103/105 passed). Ninety-nine items were available in the evaluation split, covering 44 conditions. All 99 have gold answers that are management steps rather than diagnosis names, so “missed the red flag” and “answered incorrectly” are distinct endpoints by construction.

Annotation reliability. Three independent automated annotators of differing scale judged each item; Fleiss’ $\kappa = 0.487$, with 77/99 unanimous. No physician reviewed these items. All principal analyses are repeated on the unanimous subset as a sensitivity analysis.

D. Elicitation protocol

Every episode used a fixed structured-output schema requesting, in one response: the differential actually considered, the single can’t-miss condition actively excluded, the findings used, the committed answer, and a confidence. The schema is byte-identical across all conditions, so the elicitation of confidence never varies with the experimental factor. Responses were parsed from JSON with at most one retry; parse rate was 1.000.

Elicitation-format control. To exclude the possibility that null discrimination reflects a poorly chosen response format, four formats were compared on identical items: an integer 1–10 scale, a 0–100 probability, a four-level verbal scale, and a post-hoc self-check in which the answer and the confidence judgement are produced in separate calls.

E. Interactive protocol

In the interactive setting the model receives only the presenting information and must issue at most $T = 8$ actions, each either a question to a simulated patient

or a test order, before committing. Patient responses are produced by a separate model; test results are returned by deterministic lookup against the case file, so evidence availability cannot drift between conditions. The commitment turn uses the same fixed schema as all static conditions. The response ceiling was set well above the observed reasoning draw; truncation was logged per episode and any truncated call was automatically retried at a doubled ceiling, because a ceiling near the reasoning draw manufactures non-responses at high effort.

F. Endpoints and statistics

The primary endpoint is the area under the ROC curve for elicited confidence discriminating correct from incorrect responses. AUROC is used rather than expected calibration error because ECE conflates global bias with discriminative power: a model that shifts all its confidences downward improves ECE without gaining any ability to identify its own errors. The clinical endpoint is whether the can’t-miss diagnosis appears in the stated differential or the actively-excluded field, matched through the alias set. Decision-curve analysis reports, for each confidence threshold, the fraction of cases deferred and the endpoint rate among cases retained for autonomous handling.

All intervals are bootstrap percentile intervals over 4,000 resamples at the episode level. Difficulty matching subsamples the correct responses of the comparison set until its accuracy equals that of the target set, repeated 3,000 times; AUROC is rank-based and therefore base-rate invariant in principle, and this procedure verifies it empirically.

IV. Results

A. Discrimination is task-dependent, not a fixed model property

Fig. 1 and Table I report the primary endpoint across all conditions. Within a single model, AUROC ranges from 0.492 to 0.800. On standard-difficulty clinical items confidence is informative (0.699 and 0.703); on the hard subset of the same sources it is at chance (0.492 and 0.540). The pattern reproduces in a non-reasoning model (0.671 standard, 0.475 hard), so it does not depend on explicit reasoning machinery, and in a second vendor’s model (0.480 on the hard subset).

B. The frontier collapse is not a base-rate artifact

Because accuracy differs between the standard and hard splits, the AUROC difference could in principle reflect base rate rather than difficulty. Subsampling the standard split to the accuracy of the hard split leaves AUROC unchanged: 0.698 [0.668, 0.731] at matched accuracy 0.387, and 0.703 [0.670, 0.735] at matched accuracy 0.474, against unmatched values of 0.698 and 0.703. The collapse is a property of the items, not of the class balance.

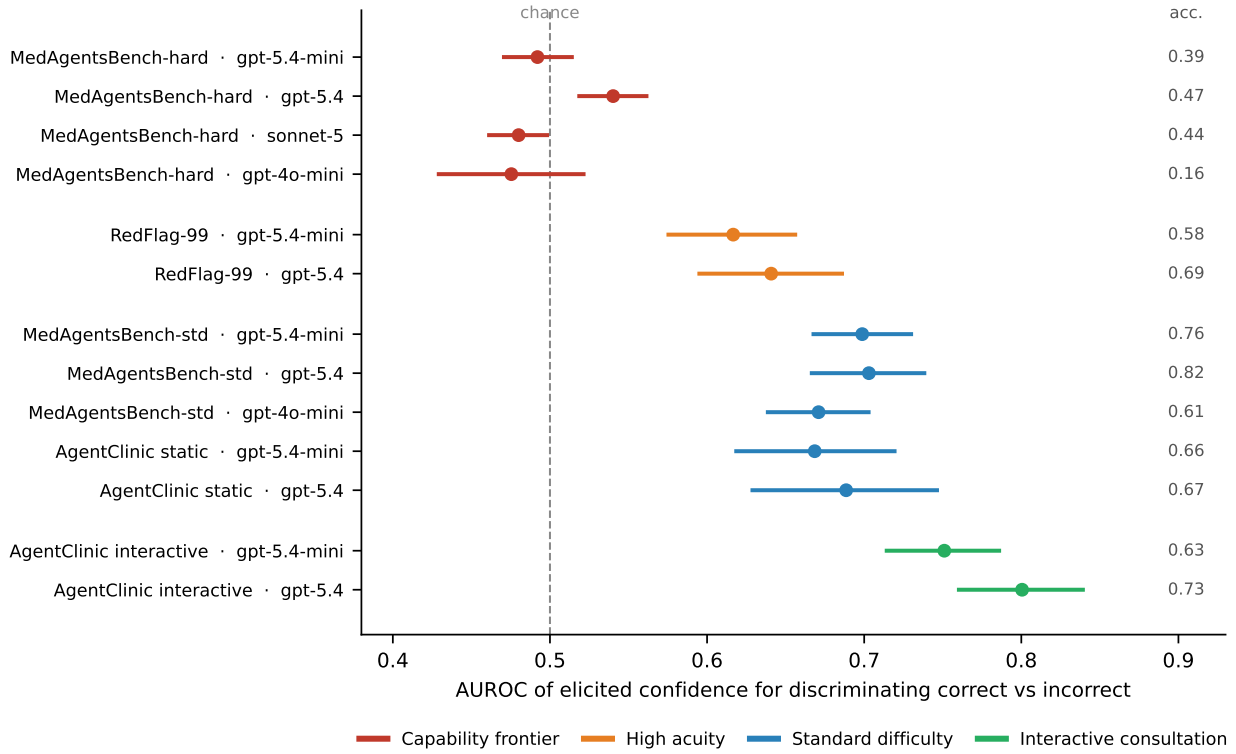


Fig. 1. Discriminative power of elicited confidence, by deployment condition. Points are AUROC with 95% bootstrap intervals over episodes; the right-hand column gives task accuracy. The same model spans chance to strong discrimination depending on condition: at the capability frontier confidence is uninformative, on high-acuity presentations it is degraded beyond what difficulty predicts, and in interactive consultation—where the model obtains its own evidence—it is most informative.

TABLE I

Discrimination of elicited confidence across conditions. AUROC with 95% bootstrap interval; * indicates the interval excludes 0.5.

Setting	Model	<i>n</i>	Acc.	AUROC [95% CI]
Clinical, static, capability frontier				
MedAgentsBench-hard	gpt-5.4-mini	2228	.387	.492 [.469, .515]
MedAgentsBench-hard	gpt-5.4	2235	.474	.540 [.517, .563]*
MedAgentsBench-hard	sonnet-5	2948	.445	.480 [.460, .499]
MedAgentsBench-hard	gpt-4o-mini	784	.157	.475 [.428, .523]
Clinical, static, standard difficulty				
MedAgentsBench-std	gpt-5.4-mini	1198	.760	.699 [.666, .729]*
MedAgentsBench-std	gpt-5.4	1200	.824	.703 [.666, .739]*
MedAgentsBench-std	gpt-4o-mini	800	.613	.671 [.637, .705]*
AgentClinic (static)	gpt-5.4-mini	409	.655	.669 [.617, .721]*
AgentClinic (static)	gpt-5.4	303	.667	.689 [.628, .748]*
Clinical, high acuity				
RedFlag-99	gpt-5.4-mini	594	.581	.617 [.575, .656]*
RedFlag-99	gpt-5.4	594	.694	.641 [.592, .688]*
Clinical, interactive consultation				
AgentClinic (inter.)	gpt-5.4-mini	642	.632	.751 [.713, .787]*
AgentClinic (inter.)	gpt-5.4	642	.727	.800 [.759, .840]*
Non-clinical controls				
MMLU-Pro	gpt-5.4-mini	600	.742	.704 [.659, .747]*
ARC-Challenge	gpt-5.4-mini	600	.958	.782 [.683, .874]*
MedCalc	gpt-5.4-mini	581	.690	.693 [.650, .736]*
GSM8K	gpt-5.4-mini	599	.846	.534 [.505, .566]*

C. No elicitation format recovers discrimination

Across four formats and two operating points, all eight cells lie at chance: integer 0.509/0.451, probability 0.540/0.451, verbal 0.486/0.490, post-hoc self-check 0.522/0.443. Repeated sampling is likewise uninformative: within a fixed operating point, self-consistency across three

samples discriminates no better than the verbalized value (0.500–0.562 versus 0.527–0.563). The failure is therefore not one of verbalization—there is no accessible signal being poorly reported.

D. Confidence degrades further on can't-miss presentations

On RedFlag-99, AUROC is 0.617 and 0.641. Against accuracy-matched controls drawn from standard-difficulty items, this is a decrement of -0.082 and -0.063 , with the RedFlag value falling outside the matched-control interval in both models (0.699 [0.681, 0.717]; 0.704 [0.688, 0.719]). Restricting to the 77 unanimously annotated items leaves the result unchanged (0.619 and 0.643; decrements -0.079 and -0.060), so the finding does not rest on items where annotators disagreed.

E. Confidence does not predict the clinical endpoint

The models failed to place the can't-miss diagnosis in the differential in 46.5% and 39.4% of red-flag episodes. Confidence discriminates this endpoint at 0.532 [0.49, 0.57] and 0.567 [0.52, 0.61]—weaker than its discrimination of answer correctness on the same episodes. Restricting to episodes with confidence ≥ 9 leaves the missed-red-flag rate at 43.2% and 36.2%, against base rates of 46.5% and 39.4%.

TABLE II

Decision-curve analysis on RedFlag-99. Deferral routes cases below the threshold to a clinician; the endpoint is the missed-red-flag rate among cases retained for autonomous handling.

Model	Threshold	Deferred	Missed rate	Δ vs. none
gpt-5.4-mini	none	0.0%	.465	—
	≥ 9	22.4%	.432	−.033
	≥ 10	74.4%	.474	+0.009
gpt-5.4	none	0.0%	.394	—
	≥ 9	22.4%	.362	−.032
	≥ 10	59.1%	.346	−.048

TABLE III

Deferral value in interactive consultation, AgentClinic (214 cases $\times$ 3 seeds).

Model	Threshold	Deferred	Acc. retained	Δ
gpt-5.4-mini	none	0.0%	.632	—
	≥ 8	18.5%	.715	+0.083
	≥ 9	37.5%	.786	+0.153
	≥ 10	74.9%	.857	+0.225
gpt-5.4	none	0.0%	.727	—
	≥ 8	24.8%	.859	+0.132
	≥ 9	44.5%	.896	+0.169
	≥ 10	71.5%	.907	+0.180

Decision-curve analysis makes the consequence explicit (Fig. 2, left; Table II). Deferring the least-confident 22.4% of red-flag cases reduces the missed-red-flag rate among retained cases by 3.3 and 3.2 percentage points. For the weaker model, raising the threshold further inverts the benefit: deferring 74.4% of cases leaves the retained set worse than no gating at all.

F. Interactive consultation restores discrimination

On the same 214 AgentClinic cases, presented interactively rather than as complete vignettes, confidence discriminates at 0.751 [0.713, 0.787] and 0.800 [0.759, 0.840], against 0.669 and 0.689 for the static presentation of the same cases. Deferral becomes useful: routing the least-confident 18.5% raises accuracy of the retained set from 0.632 to 0.715, and routing 37.5% raises it to 0.786 (Fig. 2, right; Table III). The stronger model shows the same pattern, from 0.727 to 0.859 at 24.8% deferral.

G. What elicited confidence tracks

Confidence agrees across models more strongly than accuracy does: mean pairwise correlation of elicited confidence is $r = 0.749$ across four models, against $r = 0.646$ for correctness on the same items. Within model, confidence correlates with the stability of the model’s own answer across operating points ($r = 0.36$ – 0.50), and—critically—predicts equally well from another model’s answer stability ($r = 0.30$ – 0.34), which rules out a within-episode circularity. Stability itself carries little information about correctness ($r = 0.05$ – 0.13). Confidence is therefore a readout of answer convergence on an item, and convergence is only weakly related to being right.

V. Discussion

The practical conclusion is that “is elicited confidence reliable?” has no model-level answer. The same model, on clinically similar material, spans chance to strong discrimination depending on whether the item is at its capability frontier, whether it carries high clinical acuity, and whether it gathered the evidence itself. Reporting a single calibration figure for a clinical LLM is therefore not meaningful for deployment.

Two consequences follow for system design. First, confidence-gated deferral cannot be validated on standard benchmark items and then deployed against high-acuity presentations: on exactly the cases that motivate the safety mechanism, gating recovers three percentage points of missed red flags for a 22% clinician workload, and for a weaker model an aggressive threshold makes the retained set worse. Second, and in the opposite direction, static benchmark evaluation understates confidence reliability for consultation agents. The interactive result is not a contradiction of the static one: if confidence tracks answer convergence, then a model that has actively obtained the discriminating finding converges for a reason, and its confidence inherits that reason. Evidence acquisition supplies information that a fully-specified vignette never requires the model to earn.

That mechanism also explains why no elicitation format helps. All four formats read out the same underlying convergence signal, so rewording the question cannot manufacture information that the computation did not produce. It is consistent with general-domain findings that cross-model confidence is dominated by a shared item factor [14] and that overconfidence stems from insufficient conditioning on the model’s own answer [6].

A. Limitations

No physician reviewed the RedFlag-99 items; construct validity rests on a pre-registered mechanical filter, automated annotation with reported reliability ($\kappa = 0.487$), and a sensitivity analysis on the unanimous subset. RedFlag-99 draws disproportionately from one source benchmark. The high-acuity condition was not evaluated interactively at adequate power, because only 13 of the 214 AgentClinic vignettes carry a can’t-miss diagnosis; that cell is reported descriptively and should not be over-read. Confidence was elicited jointly with the answer in a single response for the principal conditions, which is the deployment-realistic protocol but confounds elicitation with generation; the post-hoc format controls for this and shows no difference. One control dataset, GSM8K, is an outlier that we cannot explain: accuracy is high (0.846) but discrimination is near chance (0.534). We tested and rejected two accounts—answer format, since the free-response MedCalc set discriminates normally (0.693), and error type, since confidence discriminates execution slips and formula errors comparably (difference +0.043, CI [−0.034, +0.119]). A reported confound in which reasoning models re-solve the problem when asked

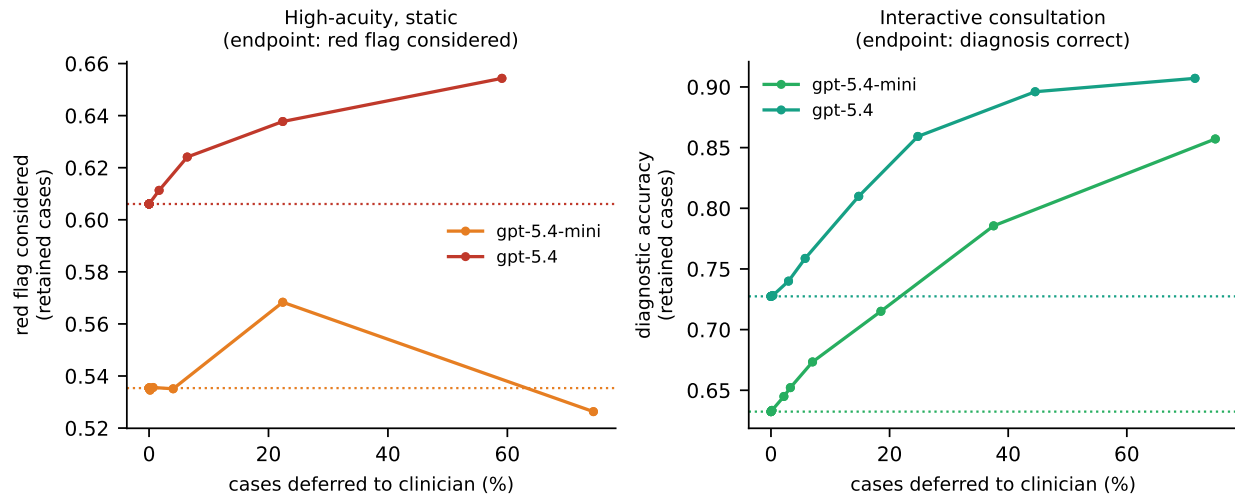


Fig. 2. Decision curves for confidence-gated deferral. Left: high-acuity static cases, endpoint “can’t-miss diagnosis considered”. Right: interactive consultation, endpoint diagnostic accuracy. Dotted lines mark the no-deferral baseline. Note the order-of-magnitude difference in vertical scale: on high-acuity cases deferral buys a few points and, for the weaker model, becomes harmful beyond 60% deferral; in interactive consultation the same mechanism on the same case material yields large, monotone gains.

about confidence [14] is a candidate we cannot test without access to reasoning traces.

VI. Conclusion

Elicited confidence in clinical LLMs is not a stable property that can be measured once and relied upon. It discriminates well on routine items and in interactive consultation, and fails at the capability frontier and—beyond what difficulty predicts—on presentations carrying can’t-miss diagnoses, where it is nearly uninformative about whether the critical diagnosis was considered at all. Deployments that gate on confidence should validate that gate in the modality and on the acuity stratum in which it will operate; a gate validated on standard benchmark questions provides no assurance for high-acuity cases.

Data and Code Availability

All benchmarks are publicly available. Episode-level records, the RedFlag-99 probe with its construction pipeline and annotation, and analysis code are released with the paper.

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